

Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK 9

ATOMIC ELECTRON STRUCTURE, TOTAL ANGULAR MOMENTUM

• **Electronic ground states / Ex. 8.2**

For all elements in the first two *periods*, list the electron descriptions in their ground state using $n\ell$ notation.

Solution

H: $1s^1$, He: $1s^2$,

Li: $1s^2 2s^1$, Be: $1s^2 2s^2$, B: $1s^2 2s^2 2p^1$, C: $1s^2 2s^2 2p^2$, N: $1s^2 2s^2 2p^3$, O: $1s^2 2s^2 2p^4$, F: $1s^2 2s^2 2p^5$, Ne: $1s^2 2s^2 2p^6$

Elements like He and Ne, characterized by a full outer shell, are *inert gases*, elements like Li, with a single electron in the outer shell are *alkali metals*, elements with a single electron shy of a full shell are *halogens*.

Comment

An alternative notation, useful for heavier elements, lists the heaviest inert gas in bracket, and list only the additional electrons in the outer shells, for instance

Li: $[\text{He}]2s^1$, Na: $[\text{Ne}]3s^1$, K: $[\text{Ar}]4s^1$, ... (*alkalis*)

Sc: $[\text{Ar}]2s^2 3d^1$, Ti: $[\text{Ar}]2s^2 3d^2$, V: $[\text{Ar}]2s^2 3d^3$, ... (*transition metals*)

• **Shells & subshells / Ex. 8.3**

How many subshells are in the following shells: K, L, N, M, and O?

Solution

The shells correspond to different values of the principal quantum number n , and the subshells correspond to the different values of orbital quantum number $\ell < n$. Therefore

- K: $n = 1$ so that $\ell = 0$ (1s)
- L: $n = 2$ so that $\ell = 0$ (2s), $\ell = 1$ (2p)
- M: $n = 3$ so that $\ell = 0$ (3s), $\ell = 1$ (3p), $\ell = 2$ (3d)
- N: $n = 4$ so that $\ell = 0$ (4s), $\ell = 1$ (4p), $\ell = 2$ (4d), $\ell = 3$ (4f)
- O: $n = 5$ so that $\ell = 0$ (5s), $\ell = 1$ (5p), $\ell = 2$ (5d), $\ell = 3$ (5f), $\ell = 4$ (5g)

• **Total Angular Momentum / Ex. 8.12**

The zirconium atom ground state has the electronic structure $[\text{Kr}]5s^2 4d^2$. (a) What are the values of the total S and L ? What are the permissible values of J ? (b) Write the spectroscopic notation for these possible values of S , L , and J . (c) Which one of these is likely to represent the ground state?

Solution

Zirconium has the electronic structure $[\text{Kr}]5s^2 4d^2$, we are concerned with the two d electrons in the open shell. The 5s shell is closed, and there are two electrons in the (open) 4d shell. According to Hund's *first* rule, the two electrons are aligned in order to maximize the total spin ($S = 1$) but due to Pauli's exclusion principle, the two electrons must occupy two different d orbitals. According to Hund's *second* rule, placing the electrons in the $m_\ell = 2$ and $m_\ell = 1$ orbitals maximized the orbital angular momentum ($L = 3$).

- (a) Then, the possible value of total angular momentum are $|L - S| \leq J \leq |L + S|$, and in this specific case $J = \{2, 3, 4\}$. Since the $L = 4$, then $n = 4$.
- (b) In the spectroscopic notation: $n^{2S+1}L_J$, the corresponding levels are denoted by 4^3F_2 , 4^3F_3 , and 4^3F_4 .
- (c) According to Hund's *third* rule, the ground state is the one with the lowest J : 4^3F_2 , because Zr outer shell is less than half-filled. If the outer shell is half-filled or more than half-filled (yet not full) the ground state is the one with the highest J instead.

• **Elements with no ground-state spin-orbit splitting / Ex. 8.14**

List all the elements through calcium (Ca) that you would expect not to have a spin-orbit interaction that splits the ground-state energy. Explain.

Solution

The term corresponding to the spin-orbit interaction (SOI) is given by

$$H_{\text{SOI}} = \xi \mathbf{L} \cdot \mathbf{S}$$

where ξ is the coupling constant. Therefore, no spin-orbit interaction requires either

(i) a **closed subshell** ($\mathbf{S} = \mathbf{0}$, since each electron has a partner with opposite spin) which occurs in

- He: $1s^2$
- Be: $[\text{He}]2s^2$
- Ne: $[\text{He}]2s^22p^6$
- Mg: $[\text{Ne}]2s^2$
- Ar: $[\text{Ne}]3s^23p^6$
- Ca: $[\text{Ar}]4s^2$

or

(ii) $\mathbf{L} = \mathbf{0}$, which we have in element with an outer (partially filled, $S = 1/2$) s shell

- H: $1s^1$
- Li: $[\text{He}]2s^1$
- Na: $[\text{Ne}]3s^1$
- K: $[\text{Ar}]4s^1$

- **Electronic structure of boron**

Boron (B) has the ground state electron configuration $[\text{He}]2s^22p^1$. Find the total S , L , J quantum numbers of the ground state and the first excited state of the B atom.

Solution

Boron is in an open-shell configuration, with a single spin in the $2p$ orbital. Therefore, $S = 1/2$ and the lowest-energy state has $L = 1$ (II Hund's rule). The total angular momentum can take values $1/2 = |L - S| \leq J \leq |L + S| = 3/2$, corresponding to $^2P_{1/2}$ (2-fold degenerate) and $^2P_{3/2}$ (4-fold degenerate). Since the $2p$ shell is less-than-half-filled, the state $^2P_{1/2}$ is lower in energy than $^2P_{3/2}$ (III Hund's rule). Because spin-orbit interaction is the smallest energy scale, the $^2P_{3/2}$ is the first excited state.

- **Electronic structure of Neodymium**

Neodymium (Nd) has the electronic configuration $[\text{Xe}] 4f^4 6s^2$. Find the total S , L , J quantum numbers of the ground state of the Nd atom. Label the ground state using the spectroscopic notation.

Solution Neodymium has an open-shell configuration with four electrons in the $4f$ shell. f orbitals have orbital angular momentum $l = 3$, and the z component of the orbital angular momentum can take 7 different values $m_\ell = -3, \dots, +3$. Therefore, in the ground state, all 4 electrons have up spins with spin quantum number $s = 1/2$, resulting in total spin $S = 4 \times \frac{1}{2} = 2$ that is maximized (I Hund's rule). The four electrons occupy four f orbitals with orbital quantum numbers $\{m_\ell\} = \{3, 2, 1, 0\}$ so the total $L = 3 + 2 + 1 + 0 = 6$ is maximized (II Hund's rule). The total angular momentum can take values $4 = |L - S| \leq J \leq |L + S| = 8$, which is minimized (III Hund's rule), so $J = 4$. The ground state corresponds to 5I_4 in the spectroscopic notation (and is $2J + 1 = 9$ -fold degenerate).

SPIN

• Stern–Gerlach splitting of Ag particles / Ex. 7.26

The magnetic field in a Stern–Gerlach experiment varies along the vertical direction as $dB_z/dz = 20.0 \text{ T/cm}$. The horizontal length of the magnet is $L = 7.10 \text{ cm}$, and the speed of the Ag atoms averages $v_x = 925 \text{ m/s}$. The average mass of the silver atoms is $m_{\text{Ag}} = 1.81 \times 10^{-25} \text{ kg}$. (a) Show that the z component of its magnetic moment is one Bohr magneton μ_B . (b) What is the separation of the two Ag atom beams as they leave the magnet?

Solution

(a) The spin magnetic moment is given by

$$\mu_s = \frac{1}{\hbar} 2\mu_B |\vec{S}|$$

where $|\vec{S}|$ indicates the length of the Ag intrinsic spin angular momentum vector and $g_s = 2$ is the gyromagnetic ratio. The electronic structure of silver is given by $[\text{Kr}]4d^{10}5s^1$, where $[\text{Kr}]$ denotes the completely filled shells of the noble gas Krypton, on top of which there is a full $4d$ subshell and a single $5s$ electron. Full subshells have zero angular momentum (both orbital and spin), and the s electron has zero orbital angular momentum. Hence all the angular momentum, and consequently, the magnetic moment, comes from the spin of the $5s$ electron.

Since the electron has a spin quantum number $s = 1/2$ and projections $m_s = \pm 1/2$, one has that

$$|\vec{S}| = \hbar\sqrt{s(s+1)} = \hbar\sqrt{3/4} \quad (1)$$

yielding

$$\mu_s = g_s \mu_B |\vec{S}| = \sqrt{3} \mu_B,$$

where we used the approximate electron gyromagnetic ratio (g -factor) $g_s = 2$. Then the μ_z component is obtained by projecting onto the z axis, with the angle given by

$$\cos(\theta) = \frac{m_s}{|\vec{S}|} = \frac{1/2}{\sqrt{3/4}} = \frac{1}{\sqrt{3}}, \quad (2)$$

so that

$$\mu_z = \mu_s \cos(\theta) = \mu_B.$$

(b) When passing through the magnet, with the field directed towards the z axis, the Ag atoms are subject to a potential

$$V = -\vec{\mu} \cdot \vec{B} = -\mu_z B_z$$

corresponding to an acceleration

$$a_z = \frac{1}{m_{\text{Ag}}} F_z = -\frac{1}{m_{\text{Ag}}} \frac{dV}{dz} = \frac{1}{m_{\text{Ag}}} \mu_z \frac{dB_z}{dz}.$$

The vertical deflection of the Ag atoms, due to the *constant* acceleration created by the constant magnetic field gradient (as the field lines become *denser* from the North to the South pole, the field is stronger) is given by

$$z = \frac{1}{2} a_z t^2$$

where the time spent inside the magnet is determined by the atoms' velocity along the x direction, i.e. $t = L/v_x$, thus yielding a deflection

$$z = \frac{1}{2} \left(\frac{1}{m_{\text{Ag}}} \mu_z \frac{dB_z}{dz} \right) \left(\frac{L}{v_x} \right)^2 = 3.017 \times 10^{-4} \text{ m},$$

obtained converting all data to SI units, i.e., $L = 0.0071$ m, $dB_z/dz = 2000.0$ T/m, and $\mu_B = 9.27 \times 10^{-24}$ J/T. As the Ag atoms with opposite spin quantum numbers are deflected in opposite directions, i.e., $\mu_z \propto m_s$, the separation between the two Ag beams is twice this amount

$$\Delta z = 2|z| = 6.035 \times 10^{-4} \text{ m.}$$

Comment: Lorentz force

Note that Ag atoms are *not* charged (unlike electrons) and do not experience a Lorentz force. If, e.g., the Ag beam was partially ionized, the Lorentz force exerted on the Ag^+ particles in this setup result in a deflection along the y direction

$$\vec{F}_L = q \vec{v} \times \vec{B} = -|q|v_x B_z \hat{y}.$$

Moreover, in this case, the Lorentz force would not cause a twofold splitting because all ionized atoms are positively charged, and thus deflected in the same direction.

Comment: twofold splitting for Ag

Beside the spin, the angular magnetic moment, associated with the quantum number ℓ , is also sensitive to a magnetic-field and result into a splitting of the different m_ℓ components. However, in the original Stern-Gerlach experiment, only a *twofold* splitting of the Ag beam was observed, suggesting a half-integer value $s = 1/2$ resulting is $2s + 1 = 2$ components.

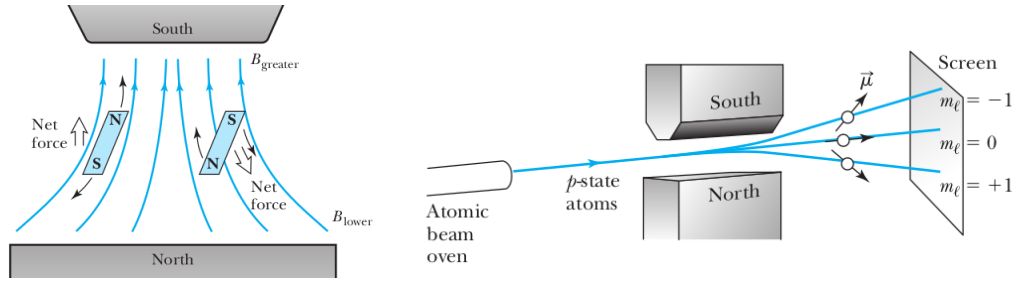


Figure 1: Setup of a Stern-Gerlach experiment. (a) The magnet poles are arranged to produce a magnetic field gradient, resulting in a net force. (b) Three patterns of atoms, due to $m_\ell = 0, \pm 1$ are expected to be observed on the screen if atoms in a p state are used. *Note: this purely-orbital splitting actually does not happen in practice, because of the electron spin.*

• **Stern-Gerlach experiment with a twist**

In the Stern-Gerlach experiment, a beam of neutral silver atoms ($S = J = 1/2$, $L = 0$) are passed through a magnetic field gradient in the z direction, which splits the beam according to the spin z component.

Now consider a setup, where the magnetic field gradient is rotated to point in the $\hat{n} = (0, 1/2, \sqrt{3}/2)$ direction ($\hat{n}$ is a normalized vector pointing 30° from the z axis in the yz plane). Now the silver atoms are split according to their eigenvalues with respect to the operator

$$S_{\hat{n}} = \hat{n} \cdot \vec{S} = \frac{1}{2}S_y + \frac{\sqrt{3}}{2}S_z,$$

where

$$\hat{S}_x = \frac{\hbar}{2}\hat{\sigma}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{S}_y = \frac{\hbar}{2}\hat{\sigma}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{S}_z = \frac{\hbar}{2}\hat{\sigma}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- (a) Calculate the normalized eigenstates of $S_{\hat{n}}$. The two states (denoted $|\uparrow\rangle$ and $|\downarrow\rangle$) with $\pm\hbar/2$ eigenvalues correspond to atomic spins polarized in the $\pm\hat{n}$ direction.

We direct the beam with $+\hat{n}$ polarization ($|\uparrow\rangle$ with $+\hbar/2$ eigenvalue) into another Stern-Gerlach setup, with the field gradient pointing in the z direction.

- (b) What are the intensities of the two beams observed after the second setup relative to the intensity of the incoming beam?

- (c) Calculate the expectation value $\langle \uparrow | S_z | \uparrow \rangle$. How is it related to the relative intensities in the previous part?

Solution

- (a) The operator is

$$S_{\hat{n}} = \frac{\hbar}{2} \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{i}{2} \\ \frac{i}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix},$$

standard calculation shows that its eigenvalues are $\pm\hbar/2$ and its unnormalized eigenvectors are

$$|\uparrow\rangle = N_{\uparrow} \begin{pmatrix} 1 \\ i(2 - \sqrt{3}) \end{pmatrix}, \quad |\downarrow\rangle = N_{\downarrow} \begin{pmatrix} 1 \\ -i(2 + \sqrt{3}) \end{pmatrix},$$

where we left the normalization constants undetermined. In order to find the normalization constants, we calculate the norm of the vectors (taking the sum of the absolute squares of the components) and demand that they are 1:

$$\langle \uparrow | \uparrow \rangle = N_{\uparrow}^2 \left(1^2 + (2 - \sqrt{3})^2 \right) = N_{\uparrow}^2 4 (2 - \sqrt{3}) = 1 \quad \Rightarrow \quad N_{\uparrow} = \frac{1}{2\sqrt{2 - \sqrt{3}}},$$

$$\langle \downarrow | \downarrow \rangle = N_{\downarrow}^2 \left(1^2 + (2 + \sqrt{3})^2 \right) = N_{\downarrow}^2 4 (2 + \sqrt{3}) = 1 \quad \Rightarrow \quad N_{\downarrow} = \frac{1}{2\sqrt{2 + \sqrt{3}}}$$

- (b) The second setup splits the beam with respect to the S_z component. We wrote the eigenstate $|\uparrow\rangle$ in the S_z basis, the two components of the wavefunction correspond to the amplitude of the state in the $+z$ and $-z$ directions. The observed intensities of the outgoing beams are proportional to the probabilities of finding the spins pointing in the $+z$ and $-z$ direction, which are given by the absolute square of the wavefunction component:

$$P(+z) = |N_{\uparrow} \times 1|^2 = \frac{1}{4(2 - \sqrt{3})} \approx 0.933,$$

$$P(-z) = |N_{\uparrow} \times i(2 - \sqrt{3})|^2 = \frac{(2 - \sqrt{3})}{4} \approx 0.067.$$

We can check that the total probability $P(+z) + P(-z) = 1$, which is a consequence of the wavefunction normalization.

- (c) To calculate the expectation value, first we act on the state on the right with the operator:

$$S_z |\uparrow\rangle = N_{\uparrow} \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ i(2 - \sqrt{3}) \end{pmatrix} = N_{\uparrow} \frac{\hbar}{2} \begin{pmatrix} 1 \\ -i(2 - \sqrt{3}) \end{pmatrix},$$

then we take the scalar product of the result with the state on the left (don't forget that the scalar product involves conjugating the "bra" state on the left)

$$\langle \uparrow | S_z | \uparrow \rangle = N_{\uparrow}^2 \frac{\hbar}{2} (1, -i(2 - \sqrt{3})) \cdot \begin{pmatrix} 1 \\ -i(2 - \sqrt{3}) \end{pmatrix} = \frac{1}{4(2 - \sqrt{3})} \frac{\hbar}{2} (1 - (2 - \sqrt{3})^2) \approx 0.866 \times \frac{\hbar}{2}.$$

This is the same result as the one that we get, if we think of the spin z component probabilistically, and use the probabilities from the previous part to calculate its expectation value:

$$\langle S_z \rangle = P(+z) \frac{\hbar}{2} + P(-z) \left(-\frac{\hbar}{2} \right) \approx 0.866 \times \frac{\hbar}{2}.$$

VARIATIONAL PRINCIPLE

• Particle in quartic potential

Consider an electron in a one-dimensional quartic potential, described by the Hamiltonian

$$H = K + V = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \lambda x^4.$$

where $\lambda = 10 \text{ eV nm}^{-4}$. Consider a variational wavefunction, characterized by a single parameter α with nm^{-2} units, and a normalization constant N :

$$\psi(x) = N e^{-\frac{1}{2}\alpha x^2}.$$

- (a) Write the normalization condition and show that it is fulfilled for $N = \left(\frac{\alpha}{\pi}\right)^{1/4}$.
Hint: use the integral identity $\int_{-\infty}^{\infty} dx e^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}}$.
- (b) Calculate the expression for the kinetic energy expectation value $\langle K \rangle_{\psi}$ that depends parametrically on α .
Hint: use the integral identity $\int_{-\infty}^{\infty} dx x^2 e^{-\alpha x^2} = \frac{1}{2} \frac{\sqrt{\pi}}{\alpha^{3/2}}$, and don't forget the normalization constant.
- (c) Calculate the expression for the potential energy expectation value $\langle V \rangle_{\psi}$ that depends parametrically on α .
Hint: use the integral identity $\int_{-\infty}^{\infty} dx x^4 e^{-\alpha x^2} = \frac{3}{4} \frac{\sqrt{\pi}}{\alpha^{5/2}}$.
- (d) Calculate the expression for the energy expectation value $E(\alpha) = \langle H \rangle_{\psi} = \langle K \rangle_{\psi} + \langle V \rangle_{\psi}$, and apply the variational principle to find an equation for α in the ground state.
- (e) Find the approximate solution α_0 and calculate its numerical value.
- (f) Calculate the approximate ground state energy $E_0 = E(\alpha_0)$ and verify that $E_0 > E'_0 = 0.2586 \text{ eV}$ (numerical solution).

Solution

- (a) The normalization condition reads

$$1 = \int_{-\infty}^{\infty} dx \psi^*(x) \psi(x) = N^2 \int_{-\infty}^{\infty} dx e^{-\alpha x^2} = N^2 \left(\frac{\pi}{\alpha}\right)^{1/2}$$

thus yielding

$$N = \left(\frac{\alpha}{\pi}\right)^{1/4}$$

- (b) The expectation value of the kinetic energy is given by

$$\langle \psi | K | \psi \rangle = -\frac{\hbar^2}{2m} \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} dx e^{-\alpha x^2/2} \left(\frac{\partial^2}{\partial x^2} e^{-\alpha x^2/2}\right) = -\frac{\hbar^2}{2m} \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} dx e^{-\alpha x^2} [-\alpha(1 - \alpha x^2)] = \frac{\hbar^2}{4m} \alpha$$

- (c) The expectation value of the potential energy is given by

$$\langle \psi | V | \psi \rangle = \lambda \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} dx x^4 e^{-\alpha x^2} = \frac{3}{4} \lambda \frac{1}{\alpha^2}$$

- (d) The parametric expression for ground state energy reads

$$E(\alpha) = \frac{\hbar^2}{4m} \alpha + \frac{3}{4} \lambda \frac{1}{\alpha^2},$$

for which one can verify that each term has the proper physical dimensions.

The variational principle implies that

$$\frac{d}{d\alpha} E(\alpha) = \frac{\hbar^2}{4m} - \frac{3}{2} \lambda \frac{1}{\alpha^3} = 0$$

which admits the solution

$$\alpha_0 = \left(\frac{6m\lambda}{\hbar^2}\right)^{1/3}.$$

(e) For the value of λ provided above, one gets

$$\alpha_0 = 9.23 \text{ nm}.$$

(f) Substituting the optimal value $\alpha = \alpha_0$ in the parametric expression for the ground state energy, one finds

$$E_0 = E(\alpha_0) = 0.26385 \text{ eV} > 0.2586 \text{ eV}.$$

• Stark effect of electron in a box

The Stark effect is the shift of atomic spectral lines in the presence of an electric field, it is analogous to the Zeeman effect which happens when a magnetic field is applied. Here we consider a simpler problem, and calculate the ground-state energy of an electron in a 1D infinite potential well when external electric field is applied. This is a good model for a nanowire in longitudinal electric field.

Consider an electron moving in the potential

$$V(x) = \begin{cases} \infty & \text{if } x > L/2 \text{ or } x < -L/2, \\ \mathcal{E}x & \text{if } -L/2 < x < L/2, \end{cases}$$

where $L = 1 \text{ nm}$ is the length of the box, and $\mathcal{E} = 0.5 \text{ eV/nm}$ is the electric field. Note that we centered the box around $x = 0$, this is so the average potential in the box remains zero.

We use a trial wavefunction which is a linear combination of the ground-state and first excited state of the box when $\mathcal{E} = 0$:

$$\psi(x) = \alpha\psi_1(x) + \beta\psi_2(x), \quad \psi_1(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right), \quad \psi_2(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi x}{L}\right).$$

(a) Using that $\psi_n(x)$ are orthonormal, find the condition on α and β that makes $\psi(x)$ normalized.

(b) Calculate the expression for the kinetic energy expectation value $\langle K \rangle_\psi$ that depends parametrically on α and β .

(c) Calculate the expression for the potential energy expectation value $\langle V \rangle_\psi$ that depends parametrically on α and β .

Hint: use the integral identity $\int_{-\pi}^{\pi} dx x \sin x \cos \frac{x}{2} = \frac{32}{9}$, and use the even/odd property of the integrands.

(d) Calculate the expression for the energy expectation value $E(\alpha, \beta) = \langle H \rangle_\psi = \langle K \rangle_\psi + \langle V \rangle_\psi$, and apply the variational principle to find an equation for α and β in the ground state.

Hint: Write the energy in the matrix form as $E = v^\dagger H v$ where $v = (\alpha, \beta)^T$ is the column vector formed of the unknown coefficients, and H is a 2×2 matrix. It is true in general, that such a function takes its minimum among normalized vectors v , when v is an eigenvector of H corresponding to its smallest eigenvalue.

(e) Calculate the variational ground state energy E_0 , and compare it with the numerical value $E'_0 = 0.36882 \text{ eV}$.

Solution

Using the Dirac notation, our trial wavefunction is $|\psi\rangle = \alpha|\psi_1\rangle + \beta|\psi_2\rangle$.

(a) Orthonormality means $\langle\psi_n|\psi_m\rangle = \delta_{nm}$, so

$$1 = \langle\psi|\psi\rangle = |\alpha|^2 \langle\psi_1|\psi_1\rangle + \alpha^*\beta \langle\psi_1|\psi_2\rangle + \alpha\beta^* \langle\psi_2|\psi_1\rangle + |\beta|^2 \langle\psi_2|\psi_2\rangle = |\alpha|^2 + |\beta|^2.$$

(b) The expectation value of the kinetic energy is written as

$$\langle\psi|K|\psi\rangle = |\alpha|^2 \langle\psi_1|K|\psi_1\rangle + \alpha^*\beta \langle\psi_1|K|\psi_2\rangle + \alpha\beta^* \langle\psi_2|K|\psi_1\rangle + |\beta|^2 \langle\psi_2|K|\psi_2\rangle = |\alpha|^2 K_{11} + |\beta|^2 K_{22},$$

where we introduced $K_{nm} = \langle\psi_n|K|\psi_m\rangle$ to denote the matrix elements, and we find using that $|\psi_n\rangle$ are eigenstates of the kinetic energy operator that $K_{11} = \frac{\pi^2 \hbar^2}{2mL^2}$ and $K_{22} = 4\frac{\pi^2 \hbar^2}{2mL^2}$.

(c) Similarly, the potential expectation value is expressed with its matrix elements $V_{nm} = \langle \psi_n | V | \psi_m \rangle$:

$$\langle \psi | V | \psi \rangle = |\alpha|^2 V_{11} + \alpha^* \beta V_{12} + \alpha \beta^* V_{21} + |\beta|^2 V_{22}.$$

Using the parity of the wavefunctions and x , we find that $V_{11} = V_{22} = 0$, and

$$V_{12} = \langle \psi_1 | V | \psi_2 \rangle = \int_{-L/2}^{L/2} dx \psi_1^*(x) \mathcal{E} x \psi_2(x) = \frac{2\mathcal{E}}{L} \int_{-L/2}^{L/2} dx x \cos\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{L}\right) = \mathcal{E} L \left(\frac{4}{3\pi}\right)^2,$$

where we changed variables and used the integral identity provided. By the properties of scalar products, $V_{21} = V_{12}^* = V_{12}$. So we find $\langle \psi | V | \psi \rangle = (\alpha^* \beta + \alpha \beta^*) V_{12}$.

(d) Combining the above two expectation values, we can write the variational energy in a matrix form

$$E(\alpha, \beta) = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}^\dagger \begin{pmatrix} K_{11} & V_{12} \\ V_{12} & K_{22} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

In order to find the minimum energy, we find the eigenvalues E of the coefficient matrix H using the characteristic polynomial:

$$0 = \det(H - E) = (K_{11} - E)(K_{22} - E) - V_{12}^2,$$

its roots are

$$E_{\pm} = \frac{K_{11} + K_{22}}{2} \pm \frac{1}{2} \sqrt{(K_{11} - K_{22})^2 + 4V_{12}^2}.$$

We see that for $\mathcal{E} = 0$ we recover the energies K_{11} and K_{22} without electric field, and for nonzero field the energy difference between the levels increases, this is called *level repulsion*. Substituting the numerical values we find for the ground state energy $E_- = 0.36885 \text{ eV}$ which is a very good approximation of the true ground state energy. In order to find the values of α and β we can substitute back the ground state energy and solve for the corresponding normalized eigenvector of H .